

```
1 ! 3456789 123456789 223456789 323456789 423456789 523456789 623456789 723456789 823456789 923456789 023456789 123456789 223456789 32
2 program least_squares
3
4     use, intrinsic :: iso_fortran_env, only : REAL64
5
6     implicit none
7
8     integer, parameter :: rp = selected_real_kind ( REAL64 ), m = 9, n = 2
9
10    ! rank one
11    real ( kind = rp ), parameter :: one ( 1 : m ) = [ 1.0_rp, 1.0_rp, 1.0_rp, 1.0_rp, 1.0_rp, 1.0_rp, 1.0_rp, 1.0_rp, 1.0_rp ]
12    real ( kind = rp ), parameter :: x ( 1 : m ) = [ 1.0_rp, 2.0_rp, 3.0_rp, 4.0_rp, 5.0_rp, 6.0_rp, 7.0_rp, 8.0_rp, 9.0_rp ]
13    real ( kind = rp ), parameter :: T ( 1 : m ) = [ 15.6_rp, 17.5_rp, 36.6_rp, 43.8_rp, 58.2_rp, 61.6_rp, 64.2_rp, 70.4_rp, &
14    98.8_rp ]
15
16    ! scalars
17    real ( kind = rp ) :: a = 0.0_rp, b = 0.0_rp, c = 0.0_rp, d = 0.0_rp, e = 0.0_rp, det = 0.0_rp, a0 = 0.0_rp, a1 = 0.0_rp
18
19    ! inner products
20    a = dot_product ( one, one )
21    b = dot_product ( one, x )
22    c = dot_product ( x, x )
23    d = dot_product ( one, T )
24    e = dot_product ( x, T )
25    ! determinant of Gram matrix
26    det = a * c - b**2
27
28    ! equations 10.26
29    a0 = ( d * c - b * e ) / det
30    a1 = ( a * e - b * d ) / det
31
32    write ( *, * ) "particular least squares solution"
33    write ( *, * ) "intercept a0 = ", a0
34    write ( *, * ) "slope a1 = ", a1
35
36    stop 'Successful termination for "least_squares.f08".'
37
38 end program least_squares
```